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Power transfer limit calculation for multi-area interconnected power networks

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ABSTRACT

Multi-area interconnected schemes are beneficial for power networks to coordinate energy resources and have created the need to identify the critical conditions for transferring power among multiple subareas. This study proposes an improved continuation power flow (CPF) model to calculate the power transfer limits (PTL) among interconnected networks corresponding to either saddle node bifurcation (SNB) or limit-induced bifurcation (LIB). Formulations of the cooperation in multi-area automatic generation control (AGC) systems are proposed to express the power balance in each subarea and the adjustments of power exchange. The predictor-corrector scheme for the improved CPF is derived and restated. Moreover, a practical step size control method is proposed in accordance with polynomial fitting analysis with the relevant computational variables to enhance the performance and efficiency of CPF steps. Case studies based on an IEEE 5-bus, IEEE 118-bus and Polish 2383-bus test system demonstrate the efficiency of the proposed CPF model and its ability to generate reliable PTL results.

1. Introduction

For reasons such as regulation and operation problems, power systems may consist of multiple subnetworks that form a multi-area interconnected scheme [1,2]. The power source structure and load demand distribution in subareas may be complementary, which makes interconnection an economical and robust method of cooperation among multiple networks. In China, interconnected power networks among different provinces allow for the transfer of sustainable energy from remote areas to load centers far away [3]. This transfer is not only beneficial for meeting the electric power demand associated with economic development but also a significant step toward low-carbon and clean energy integration [4]. To economically and efficiently transfer renewable power from western to eastern areas [5], a transmission network framework based on ultrahigh voltage alternating current [6] and ultrahigh voltage direct current [7] is being established in China. Once fully completed, the transmission system will cover provinces across China through multi-area interconnected networks that will support a wide range of complementary power transfer schemes and energy types [8].

For such a multi-area interconnected power system, cooperation to maintain a power balance and control the power exchange is particularly important [9]. Since the power required by the rectifier and inverter in direct-current (DC) systems can be directly and precisely adjusted [10], the alternating-current (AC) power flow problems among interconnected synchronous power grids have been given extensive attention [11]. For power exchange through AC power transmission tie lines among interconnected networks, multi-area automatic generation control (AGC) systems play an important role in system operation [12]. Multi-area AGC systems should support an internal power balance within each subarea and adjust the power exchange among subareas to specified target values [13]. For power flow problems with the assumption that the system frequency is fixed, an imbalance of power mainly represents the active power loss variation due to changes in the power flow distribution. Once given base values and participation coefficients for AGC-involved generators, the multi-area AGC system can adjust the power exchange among multiple networks with a certain frequency. The operation scheme of a multi-area AGC system markedly impacts the power flow of the tie lines and power distribution among subnetworks [14]. This issue should be considered when using Newton-Raphson iterative method-based power flow calculations to analyze the system state and power flow distribution of an interconnected power

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Nomenclature

P_i	Active power injection at bus *i*.
Q_i	Reactive power injection at bus *i*.
f_i^P	Active power balance equation for bus *i*.
f_i^Q	Reactive power balance equation for bus *i*.
h_i^P	Formulation of active power injection at bus *i*.
h_i^Q	Formulation of reactive power injection at bus *i*.
$\mathbf{f}^P$	Active power balance equations.
$\mathbf{f}^Q$	Reactive power balance equations.
$\mathbf{F}$	Equations of the multi-area power flow model.
$\boldsymbol{\theta}$	Vector of voltage phase angles at PV and PQ buses.
θ_i	Phase angle of bus voltage V_i.
$\mathbf{V}$	Vector of voltage magnitudes at PQ buses.
V_i	Voltage magnitude at bus *i*.
$\alpha_{i,a}$	AGC-participating coefficient for bus *i* in area *a*.
$\boldsymbol{\mu}$	Vector of coefficients for unbalanced power.
μ_a	Coefficient for the unbalanced power in area *a*.
λ	Power increment parameter for CPF.
K_i^P	Active power increment coefficient for bus *i*.
K_i^Q	Reactive power increment coefficient for bus *i*.
$\mathbf{K}$	Vector of power increment coefficients.
NB	Total bus number in the whole system, ($NQ + NV$).
NQ	PQ bus number in the whole system.
NV	PV bus number in the whole system.
NA	Number of areas in the whole system.
θ_{ij}	Phase angle between complex bus voltages V_i and V_j.
G_{ii}	Self-conductance at bus *i*.
B_{ii}	Self-susceptance at bus *i*.
G_{ij}	Mutual conductance between buses i and j.
B_{ij}	Mutual susceptance between buses *i* and *j*.
Ω_a^A	Collection of tie lines and buses corresponding to area *a*.
Ω^{PV}	Collection of PV buses.
Ω^{PQ}	Collection of PQ buses.
P_l	Active power flow transmitted by the *l*th tie line.
G_l	Series conductance of the *l*th tie line.
B_l	Series susceptance of the *l*th tie line.
P_a^A	Initial value of the active power output of area *a* through interconnected network tie lines.
ΔP_a^A	Incremental of P_a^A due to the increase of unbalanced power at area *a*.
$\mathbf{X}$	Combination of $\boldsymbol{\theta}$, $\mathbf{V}$, $\boldsymbol{\mu}$ and λ.
$\mathbf{A}$	Matrix of multi-area AGC-participating coefficients.
ε_{min}	Confirmation coefficient for Newton iteration convergence.
ε_{max}	Confirmation coefficient for Newton iteration divergence.
s	Count of Newton iterations.
sc	Count of Newton iterations when convergence occurs.
sm	Maximum allowable count of Newton iterations.
$\mathbf{J}$	Jacobian matrix for the conventional power flow model.
$\mathbf{J}_{PF}$	Jacobian matrix for the multi-area power flow model.
$\mathbf{J}_{CPF}$	Jacobian matrix for the corrector in the improved CPF model.
X_τ	The τth element of vector $\mathbf{X}$.
$\mathbf{X}^{base}$	Solution at the predictor–corrector base point.
X^{pre}	Solution at a predictor–corrector prediction point.
X_τ^{pre}	The τth element of vector $\mathbf{X}^{pre}$.
σ	Step size control parameter.
σ_{min}	Minimum value of the step size control parameter.
$d\mathbf{X}$	Tangent predictor vector for CPF.
Q_i^G	Generator reactive power at bus *i*.
Q_i^M	Maximum allowable reactive power output at bus *i*.
η_i	Reactive power reserve at PV bus *i*.
γ	Coefficient of numerical deviation.
$\mathbf{L}$	Inverse matrix of $\mathbf{J}_{PF}$.
L_{ii}	Diagonal element of matrix $\mathbf{L}$ at column *i* and row *i*.
t	Number of recorded valid calculation results.

network. Multi-area AGC is treated in traditional system analysis by alternating calculations for conventional power flow and AGC generator output adjustments [15]. Dong et al. [12] presented an integrated multi-area power flow model with good convergence because all the variables are synchronously adjusted inNewton iterations.

However, when testing the power transfer limit (PTL) [16] related to saddle node bifurcation [17,18], the Newton iterative method-based power flow model generally encounters numerical ill-conditions because the Jacobian matrix tends to be singular. In such cases, a continuation method featuring the predictor–corrector scheme is introduced to establish the traditional CPF model [19,20]. The CPF model simulates synchronous increases in power generation and the power demand with specified growth patterns, and changes in the system state considering bus mode conversion caused by generator reactive power limitations are identified [21]. Particularly, when a PV mode bus [22] located in a weak voltage-supported network is forced to be a PQ mode bus [23] due to the exhaustion of the reactive power reserve [24], the system may suddenly fall into an unstable state defined by limit-induced bifurcation (LIB) [17]. Continuation power flow (CPF) theory can be effectively applied to analyze the static voltage stability of power systems [25] because of its capability to identify various critical conditions, including saddle node bifurcation (SNB) and LIB. Moreover, [21] focused on improving the CPF based on the step size control effectiveness and highlighted the effects of AGC on the PTL calculation results for isolated power networks. But it has not been taken into account that multi-area automatic generation control (AGC) systems play a role in controlling the power exchanges between different subnetworks.

Therefore, as a practical method of assessing the static voltage stability of a system, the CPF approach is expected to play an important role in analyzing power flow problems concerning multi-area interconnected power networks. In [26], the conventional CPF method considering SNB was combined with an approximate network equivalent-based method to analyze the total transfer capacity of multi-area interconnected networks. This study proposes an improved CPF model for multi-area networks with precise expressions of the power balance and AGC exchange mechanism to identify different voltage stability bifurcation conditions and develop an efficient step size control method.

Although CPF is already well developed, few papers in the literature investigate PTL of multi-area interconnected networks while considering power exchange between different subareas. To resolve such problems of PTL and voltage stability in multi-area interconnected networks, the parameterized formulations of power flow as well as the CPF model which consider power exchange and AGC is proposed in this paper. Faced with the challenge from divergence of Newton iterations, formulations of predictor and corrector of multi-area CPF model are derived. Since the low efficiency of multi-area CPF calculations when using the traditional step size control method, a practical step size control method is established in accordance with the polynomial fitting analysis.

The remainder of this paper is organized as follows. Section II presents the parameterized power flow model for multi-area connected power networks. In Section III, the formulations for the predictor–corrector scheme of the improved CPF model are derived, and the improved step size control method is discussed. Section IV presents case studies based on the IEEE 5-bus, IEEE 118-bus and Polish 2383-bus test systems to demonstrate the performance of the proposed model. In

Section V, the main results are summarized, and the key conclusions are drawn.

2. Formulation of the parameterized power flow model for multi-area interconnected networks

For multi-area interconnected power networks, AGC-participating generators must not only be involved in the active power balance between generation and the required load but also control the power exchange among subnetworks according to the specified schedule. Accordingly, the formulation for the power injected into networks is given in (1).

$$\mathbf{f}(\theta, \mathbf{V}, \mu, \lambda) = \begin{cases} \mathbf{f}^P = \begin{cases} \mathrm{f}_i^P = P_i - \mathrm{h}_i^P(\theta, \mathbf{V}) + \beta_i + \lambda K_i^P = 0 \\ (i = 1, 2, \cdots, NB) \end{cases} \\ \mathbf{f}^Q = \begin{cases} \mathrm{f}_i^Q = Q_i - \mathrm{h}_i^Q(\theta, \mathbf{V}) + \lambda K_i^Q = 0 \\ (i = 1, 2, \cdots, NQ) \end{cases} \end{cases} \tag{1}$$

With

$$\theta = [\theta_1 \quad \theta_2 \quad \cdots \quad \theta_{NB-1}] \tag{2}$$

$$\mathbf{V} = [V_1 \quad V_2 \quad \cdots \quad V_{NQ}] \tag{3}$$

$$\mu = [\mu_1 \quad \mu_2 \quad \cdots \quad \mu_{NA}] \tag{4}$$

This formulation involves the active power balance equations for all the buses and the reactive power balance equations for all the PQ buses. The detailed expressions for h_i^P (V, θ) and h_i^Q (V, θ) are given as follows.

$$h_i^P(\theta, \mathbf{V}) = \sum_{j \in i}^{n} V_i V_j (G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij}) \tag{5}$$

$$h_i^Q(\theta, \mathbf{V}) = \sum_{j \in i}^{n} V_i V_j (G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij}) \tag{6}$$

Since the conventional slack bus is eliminated and power imbalances are handled by several AGC-participating buses, the network losses and unbalanced power of each subnetwork are allocated by its generators according to the AGC participation factor. The voltage phase angle of one PV bus is specified to be the reference value, which is typically 0. In (1), the adjusted power injected into a network due by the participating AGC for each bus is indicated by the parameter β, which can be calculated in accordance with the relation in (7):

$$\begin{cases} \beta = [\beta_1 \quad \beta_2 \quad \cdots \quad \beta_{NB}]^T = \mu \mathbf{A} \\ \beta_i = \sum_{a=1}^{NA} \alpha_{i,a} \mu_a \quad (i = 1, 2, \cdots, NB) \end{cases} \tag{7}$$

where the matrix A, which is detailed in (8), is introduced to represent all the AGC-participating coefficients for all the buses. The coefficients of the buses that really support AGC power adjustment are specified as positive values, and the other coefficients are set to 0.

$$\mathbf{A} = \begin{bmatrix} \alpha_{1,1} & \alpha_{2,1} & \cdots & \alpha_{NB,1} \\ \alpha_{1,2} & \alpha_{2,2} & \cdots & \alpha_{NB,2} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{1,NA} & \alpha_{2,NA} & \cdots & \alpha_{NB,NA} \end{bmatrix} \tag{8}$$

The sum of the AGC-participating coefficients for one subarea equals 1, corresponding to the elements of one row in matrix A, which satisfies the relation in (9). If $\mu_a > 0$, the output of the AGC units in area a is upregulated to meet the unbalanced power demand, and vice versa.

$$\mu_a \sum_{i=1}^{NB} \alpha_{i,a} = \mu_a \tag{9}$$

Particularly, the introduction of power increment parameters and coefficients allows relation (1) to represent the system state corresponding to a certain additional power increase. When an assumed regular increase in the power generation, transmission and demand, λ is used to indicate the level of the power increase, and the coefficients in vector (10) specify the direction of power growth. Moreover, K should be assigned fixed values prior to power flow calculations to indicate that the generation quantity and load proportionately increase.

$$\mathbf{K} = \begin{bmatrix} K_1^P & K_2^P & \cdots & K_{NB}^P & K_1^Q & K_2^Q & \cdots & K_{NQ}^Q \end{bmatrix}^T \tag{10}$$

As shown in (11), the active power output of the multi-area AGC generators to control the power exchange can be expressed in terms of the total transmission power carried by the related tie lines among subareas. With the assumption that the lth tie line connects bus i and bus j, the detailed formula for calculating P_l in (11) can be found in (12). Note that since the branch power flow calculation results at both sides of one transmission line are different, P_l in (11) denotes the branch power flow corresponding to the bus i side of the tie line.

$$\mathbf{g}(\theta, \mathbf{V}, \lambda) = \begin{cases} g_a = \sum_{l \in \Omega_a^{\mathrm{A}}} \varphi_l P_l - (P_a^{\mathrm{A}} + \Delta P_a^{\mathrm{A}}) = 0 \\ \left(\Delta P_a^{\mathrm{A}} = \lambda \sum_{k \subset \Omega_a^{\mathrm{A}}} K_k^P; \, a = 2, 3, \cdots, NA \right) \end{cases} \tag{11}$$

With

$$P_l = V_i^2 G_l - V_i V_j (G_l \cos\theta_{ij} + B_l \sin\theta_{ij}) \tag{12}$$

$$\varphi_l = \begin{cases} 1 & (i \subset \Omega_a^{\mathrm{A}}, j \not\subset \Omega_a^{\mathrm{A}}) \\ -1 & (i \not\subset \Omega_a^{\mathrm{A}}, j \subset \Omega_a^{\mathrm{A}}) \end{cases} \tag{13}$$

where ($l \in \Omega_a^{\mathrm{A}}$) indicates that one side of the lth tie line is located at area a and ($k \subset \Omega_a^{\mathrm{A}}$) means that bus k is located at area a. Relation (11) reveals that the cumulative total power increase in area a determined by λ and K should equal the increase in the power exchange of area a to support other areas. P_a^{A} represents the base value of the exchange power of area a, for which $P_a^{\mathrm{A}} < 0$ indicates feeding power to the ath area and $P_a^{\mathrm{A}} > 0$ indicates outputting power from the ath area. The relations in (11) also indicate that the power exchanges among areas are changed along with the synchronous growth in power generation or demand depended by λ. In real power system operation, the power exchange among subareas equals the sum of the branch power flows measured at multiple terminals on the conducting tie lines. As shown in Fig. 1, the measurement points marked with "M" can be located at either side of a tie line, and the measured branch power flows are shared among subareas. In relation (13), φ_l is used to indicate the branch power measured at different sides of a tie line.

The relations for NA-1 areas but not NA areas are involved in (11) to describe the power exchange in the entire interconnected system because the sum of the power exchange in all subareas equals 0. Consequently, a compact formulation of the power flow in multi-area interconnected networks is given in (14), which includes both the power balance equations in (1) and the area power exchange relations in (11).

$$\mathbf{F}(\mathbf{X}) = \begin{cases} \mathbf{f}(\theta, \mathbf{V}, \mu, \lambda) = 0 \\ \mathbf{g}(\theta, \mathbf{V}, \lambda) = 0 \end{cases} \tag{14}$$

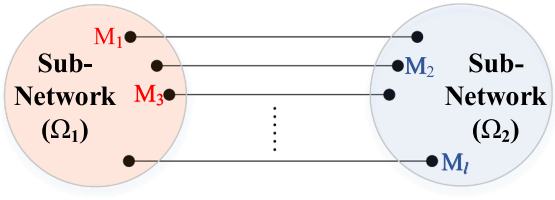


Fig. 1. Schematic diagram of interconnected network measurement points.

With

$$\mathbf{X} = [\mathbf{V}, \theta, \mu, \lambda] = [x_1 \quad x_2 \quad \cdots \quad x_{(NB+NQ+NA)}] \tag{15}$$

Since the number of unknown variables in (14) reaches $NB + NQ + NA$, one more than the number of involved equations, as shown in (15), (14) cannot be directly solved by the Newton iterative method. Notably, when λ is given, the solutions to the equations are determined corresponding to a specified system state with a certain power increment. The equations in (14), for which the solutions rely on the parameter λ, are defined here as a parameterized power flow model. This model is equivalent to the conventional power flow model when λ is set to 0. Thus, we set $\lambda = 0$ and give the formulations of the Newton iterations in (16). In this case, all of the variables can be synchronously updated step by step until they converge via Jacobian matrix-based iterations.

$$\begin{cases} \mathbf{F}(\mathbf{Y}) = [\mathbf{J}_{PF}^{(s)}][\Delta\mathbf{Y}^{(s)}]^T \\ [\mathbf{Y}^{(s+1)}] = [\mathbf{Y}^{(s)}] + [\Delta\mathbf{Y}^{(s)}] \end{cases} (\mathbf{Y} = [\theta^{(s)}, \mathbf{V}^{(s)}, \mu^{(s)}]^T) \tag{16}$$

where (s) and $(s + 1)$ represent the iteration step counter and "Δ" represents a vector consisting of the deviation values for all involved variables. Specifically, for the initial iteration indicated by $s = 0$, the unknowns, including **θ**, **V** and **μ**, should be initialized for a "flat start" according to [12]. The relation in (17) is used as the convergence standard of the Newton iteration process by replacing the variables in (14) with updated numerical solutions and testing the deviation of each equation.

$$\|\mathbf{F}(\mathbf{Y}^{(sc)})\|_\infty \leqslant \varepsilon_{\min} \tag{17}$$

where ε_{min} is a small positive number (10^{-5} in the following case studies) that is given in advance to guarantee the accuracy of the solution. In (16), $\mathbf{J}_{PF}$ represents the extended integrated Jacobian matrix of the multi-area power flow model, which is expressed as follows:

$$[\mathbf{J}_{PF}] = \left[\begin{array}{cc:c} \multicolumn{2}{c:}{\multirow{2}{*}{$\mathbf{J}$}} & \frac{\partial \mathbf{f}}{\partial \boldsymbol{\mu}} \\ & & \\ \hdashline \frac{\partial \mathbf{g}}{\partial \boldsymbol{\theta}} & \frac{\partial \mathbf{g}}{\partial \mathbf{V}} & \mathbf{0} \end{array}\right] = \left[\begin{array}{ccc} \frac{\partial \mathbf{f}}{\partial \boldsymbol{\theta}} & \frac{\partial \mathbf{f}}{\partial \mathbf{V}} & \frac{\partial \mathbf{f}}{\partial \boldsymbol{\mu}} \\ \frac{\partial \mathbf{g}}{\partial \boldsymbol{\theta}} & \frac{\partial \mathbf{g}}{\partial \mathbf{V}} & \mathbf{0} \end{array}\right] \tag{18}$$

where **J** is the conventional Jacobian matrix, which can be obtained according to [15]; **0** represents a matrix or vector in which all the elements equal zero; $\partial\mathbf{f}/\partial\boldsymbol{\theta}$ represents the matrix involving the sensitivity of each equation in **f** to different variables in vector **θ**; and the definitions of $\partial\mathbf{f}/\partial\mathbf{V}$, $\partial\mathbf{f}/\partial\boldsymbol{\mu}$, $\partial\mathbf{g}/\partial\boldsymbol{\theta}$, and $\partial\mathbf{g}/\partial\mathbf{V}$ are similar to that of $\partial\mathbf{f}/\partial\boldsymbol{\theta}$. Moreover, the elements of $\partial\mathbf{f}/\partial\boldsymbol{\mu}$ can be obtained according to the following relation.

$$\frac{\partial \beta_i}{\partial \mu_a} = \alpha_{i,a} \quad (i = 1, 2, \cdots, NB;\ a = 2, 3, \cdots, NA) \tag{19}$$

Thus, matrix $\partial\mathbf{f}/\partial\boldsymbol{\mu}$ be abbreviated as follows.

$$\frac{\partial \mathbf{f}}{\partial \mu} = \begin{bmatrix} \mathbf{A} \\ \mathbf{0} \end{bmatrix} \tag{20}$$

The elements in $\partial\mathbf{g}/\partial\boldsymbol{\theta}$ can be obtained by the following relation.

$$\frac{\partial g_a}{\partial \theta_k} = \sum_{l \in \Omega_a} \left(\frac{\partial P_l}{\partial \theta_k}\right) \quad (a = 2, 3, \cdots, NA;\ k = 2, \cdots, NB) \tag{21}$$

With

$$\frac{\partial P_l}{\partial \theta_k} = \begin{cases} V_i V_j (G_l \sin\theta_{ij} - B_l \cos\theta_{ij}) & (k = i) \\ V_i V_j (B_l \cos\theta_{ij} - G_l \sin\theta_{ij}) & (k = j) \\ 0 & (k \neq i,\ k \neq j) \end{cases} \tag{22}$$

Matrix $\partial\mathbf{g}/\partial\mathbf{V}$ can be obtained according to the following equation.

$$\frac{\partial g_a}{\partial V_k} = \sum_{l \in \Omega_a} \left(\varphi_l \frac{\partial P_l}{\partial V_k}\right) \quad (a = 2, 3, \cdots, NA;\ k = 1, 2, \cdots, NQ) \tag{23}$$

With

$$\frac{\partial P_l}{\partial V_k} = \begin{cases} 2V_i G_l - V_j (G_l \cos\theta_{ij} + B_l \sin\theta_{ij}) & (k = i) \\ -V_i (G_l \cos\theta_{ij} + B_l \sin\theta_{ij}) & (k = j) \\ 0 & (k \neq i,\ k \neq j) \end{cases} \tag{24}$$

3. Improved CPF model for Multi-area interconnected networks

When λ is specified as a certain positive value and the applicable power increment coefficients are given, the parameterized power flow model can represent any static system state. However, if the system approaches an ill-conditioned state, such as one created by SNB points, the Newton iterative method will face convergence problems that make it unable to successfully obtain the numerical power flow solutions. Thus, the predictor–corrector scheme-based continuation method is introduced to generate the improved CPF model.

3.1. CPF predictor

The predictor aims to obtain a prediction point according to the known base point generated by the Newton iterations in the power flow model or the CPF corrector. The relations in (25)–(27) detail the method used to calculate the predicted point.

$$\mathbf{X}^{pre} = \mathbf{X}^{base} + \sigma \frac{d\mathbf{X}}{\|d\mathbf{X}\|_2} \tag{25}$$

where $d\mathbf{X}$ is considered the prediction tangent vector, which can be obtained by solving (26).

$$[\mathbf{J}_{CPF}][d\mathbf{X}]^T = \begin{bmatrix} \mathbf{J}_{PF} & \mathbf{K} \\ & \mathbf{K}_a \\ \hdashline & \mathbf{e}_\tau \end{bmatrix} [d\mathbf{X}]^T = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} \tag{26}$$

With

$$\begin{cases} \mathbf{K}_a = \left[-\sum_{i \subset \Omega_2} K_i^P \quad -\sum_{i \subset \Omega_3} K_i^P \quad \cdots \quad -\sum_{i \subset \Omega_{NA}} K_i^P \right]^T \\ \mathbf{e}_\tau = [0 \quad \cdots \quad 0 \quad 1 \quad 0 \quad \cdots \quad 0] \end{cases} \tag{27}$$

In (25), the last element of $d\mathbf{X}$, $d\lambda$, equals 0 at an SNB point, and changes in this value from positive to negative can be used to confirm the ability of the predictor–corrector method to encounter the SNB point. σ is a parameter that controls the size of the predictor–corrector step by adjusting the distance between the predicted point and the base point. Additional treatments of σ considered in the improved step size control method can be found in the subsequent sections. In (27), the elements in $\mathbf{K}_a$ denote the active power increment sensitivity of each subarea as λ increases, which also causes the power exchange among areas to change. $\mathbf{e}_\tau$ represents a vector in which the τth element equals 1, and the others are zero. The parameter τ can be obtained according to the relation in (28), where the infinite norm is calculated to determine the element of $d\mathbf{X}$ with largest absolute value. In the first step of the CPF, τ can be initialized as $2NB\text{-}NV + 1$, which corresponds to the position of $d\lambda$.

$$\tau = \{\tau \mid |dX_\tau| = \|d\mathbf{X}\|_\infty\} \tag{28}$$

3.2. CPF corrector

In the case of setting the specified variable as the predicted value and supplementing corresponding relation, the equations for the corrector are established as shown in (29), in which the number of variables equals the number of equations.

$$\mathbf{G}(\mathbf{X}) = \begin{cases} \mathbf{F}(\mathbf{X}) = 0 \\ X_\tau - X_\tau^{pre} = 0 \end{cases} \tag{29}$$

Therefore, the equations in (29) can be solved using the Newton iterative method, for which the corresponding relations are shown in (30).

$$\begin{cases} \mathbf{G}(\mathbf{X}^{(s)}) = [\mathbf{J}_{CPF}^{(s)}][\Delta\mathbf{X}^{(s)}]^T \\ \mathbf{X}^{(s+1)} = \mathbf{X}^{(s)} + \Delta\mathbf{X}^{(s)} \end{cases} \tag{30}$$

where $\mathbf{X}^{(0)}$ is initialized in accordance with $\mathbf{X}^{pre}$. Relation (31) is used to confirm the convergence of the Newton iterations.

$$\|\mathbf{G}(\mathbf{X}^{(sc)})\|_\infty \leqslant \varepsilon_{min} \tag{31}$$

When the iterations converge, an updated system state representing a new level of power generation and demand is captured. Then, the number of iterations required for convergence is recorded as sc. Furthermore, the Newton iterations may encounter divergent conditions, which can be confirmed by the relation in (32). The treatment method for iteration failure will be given in the subsequent section.

$$\|\mathbf{G}(\mathbf{X}^{(sc)})\|_\infty > \varepsilon_{max} \tag{32}$$

3.3. Recognition of LIB

With sufficient reactive power support, the voltage magnitudes of some buses can be controlled as a specific value. These buses are known as PV buses. However, as λ increases to reflect the power generation and load, the reactive power generation at the PV buses may reach its upper limit, thereby forcing bus mode conversion to PQ mode.

$$\begin{cases} \eta_{min} = \min\{\eta_i \mid i \subset \Omega^{\mathrm{PV}}\} < 0 \\ \eta_i = Q_i^G - Q_i^M \end{cases} \tag{33}$$

Then, if the voltage magnitude at a converted bus becomes uncontrollable, the system state may suddenly become unstable due to the loss of significant reactive power/voltage support. Additionally, the equations in (29) must be supplemented by the corresponding reactive power balance equation. According to [21], reactive power/voltage sensitivity (the so-called VQ sensitivity) analysis for all load buses can be used to confirm this unstable system state known as LIB. Thus, the inverse matrix of $\mathbf{J}_{PF}$, as shown in (34), must be calculated.

$$\mathbf{L} = (\mathbf{J}_{PF}^{(sc)})^{-1} = \begin{bmatrix} \frac{\partial\theta}{\partial\mathbf{f}} & \frac{\partial\mathbf{V}}{\partial\mathbf{f}} & \frac{\partial\mu}{\partial\mathbf{f}} \\ \frac{\partial\theta}{\partial\mathbf{g}} & \frac{\partial\mathbf{V}}{\partial\mathbf{g}} & \mathbf{0} \end{bmatrix} \tag{34}$$

Then, some elements of $\mathbf{L}$ corresponding to the load buses are involved in (35) to confirm the existence of LIB.

$$\delta = \min\left\{ \frac{\partial\mathbf{V}_i}{\partial Q_i} \;\middle|\; i \subset \Omega^{\mathrm{PQ}} \right\} \tag{35}$$

If $\delta > 0$, the voltage magnitude increases as injected reactive power increases at each PQ bus, and the system is stable; otherwise, if there is at least one bus voltage decreases with the increase of injected reactive power, the system becomes unstable.

3.4. Controlling the step size

According to [21] the *sc* values for corrector iterations can be employed to establish a self-adaptive step size control method in which σ is adjusted at the beginning of each predictor step based on the relation in (36). As observed, after initialization with σ_{min}, a threshold value is used to control the step size, where σ is increased to accelerate the predictor–corrector steps when $sc \leq 2$ and decreased when $sc > 2$ to guarantee satisfactory numerical convergence. However, the Newton iterative method for multi-area power flow calculations requires more iterations than in other cases to coordinate and adjust the power flow among subareas; thus, the method in (36) does not perform well. Furthermore, simply increasing the threshold value leads to unacceptable divergence cases.

$$\sigma(t) = \begin{cases} \sigma_{min} \ (t = 0) \\ \sigma(t-1) \times 2 \ (t \neq 0;\ sc \leqslant 2) \\ \sigma(t-1) \div 2 \ (t \neq 0;\ sc > 2;\ \sigma > \sigma_{min}) \end{cases} \tag{36}$$

Thus, based on the relation in (36), this study presents a supplementary treatment for σ by estimating values of λ at bus mode conversion or SNB points to further improve the performance of step size control. A polynomial fitting method is used to obtain the approximate relationship between λ and the related variables. First, the generator reactive power, Q^G, for each PV bus is updated at the end of each corrector calculation step using relation (37).

$$Q_i^G = \mathrm{h}_i^Q(\theta, \mathbf{V}) - Q_i^L \quad (i \subset \Omega^{\mathrm{PV}}) \tag{37}$$

The values of V and Q^G are continually recorded along with CPF steps. Here, the valid records are generally saved in sequence unless bifurcation conditions or numerical divergence is encountered; in such cases, the records are cleared, and t is set to 0. The index in (38) is defined and calculated to ensure that the deviation in the last three valid records of either V or Q^G is sufficiently large for generating effective nonlinear fitting parameters.

$$\begin{cases} \psi_{min} = \min\{\min\{\psi_i^Q \mid i \subset \Omega^{\mathrm{PV}}\}, \min\{\psi_j^V \mid j \subset \Omega^{\mathrm{PQ}}\}\} > \gamma \\ \psi_i^Q = \min\{|Q_i^G(t) - Q_i^G(t-1)|, |Q_i^G(t-1) - Q_i^G(t-2)|\} \\ \psi_j^V = \min\{|V_j(t) - V_j(t-1)|, |V_j(t-1) - V_j(t-2)|\} \end{cases} \tag{38}$$

When relation (38) is satisfied, the relations in (39) and (40) are used to fit the approximate polynomial relationships between λ and Q_i^G. Then, λ is estimated at the point where Q_i^G reaches its upper limit Q_i^M Note that t must be reset to 0 if a PV bus is converted to PQ mode because the trends of system state variables change.

$$\lambda_i^Q = a_0 \quad (t \geqslant 3;\ \psi_{min} > \gamma;\ i \subset \Omega^{\mathrm{PV}}) \tag{39}$$

With

$$\begin{bmatrix} 1 & \eta_i^2(t) & \eta_i^4(t) \\ 1 & \eta_i^2(t-1) & \eta_i^4(t-1) \\ 1 & \eta_i^2(t-2) & \eta_i^4(t-2) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \lambda(t) \\ \lambda(t-1) \\ \lambda(t-2) \end{bmatrix} \tag{40}$$

Then, the minimum acceptable value of λ_i^Q for all the PV buses is expressed by (41).

$$\lambda_{min}^Q = \min\{\lambda_i^Q \mid i \subset \Omega^{\mathrm{PV}};\ \lambda_i^Q > \lambda(t)\} \tag{41}$$

To estimate the value of λ at an SNB point, polynomial fitting analysis between λ and each voltage magnitude at PQ buses is employed according to (42) and (43).

$$\lambda_i^V = \frac{4a_0a_2 - a_1^2}{4a_2} \quad (i \subset \Omega^{\mathrm{PQ}}) \tag{42}$$

With

$$\begin{bmatrix} 1 & V_i(t) & V_i^2(t) \\ 1 & V_i(t-1) & V_i^2(t-1) \\ 1 & V_i(t-2) & V_i^2(t-2) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \lambda(t) \\ \lambda(t-1) \\ \lambda(t-2) \end{bmatrix} \tag{43}$$

Then, the minimum estimated value of λ corresponding to SNB is expressed by (44).

$$\lambda_{min}^V = \min\{\lambda_i^V \mid i \subset \Omega^{\mathrm{PQ}};\ \lambda_i^V > \lambda(t)\} \tag{44}$$

As a result, when the constraint conditions $t \geq 3$ and $\psi_{min} > \gamma$ are both satisfied, σ is set according to the relation in (45).

$$\sigma = \frac{\min\{\lambda_{min}^V, \lambda_{min}^Q\} - \lambda(t)}{d\lambda} \quad (t \geqslant 3;\ \psi_{min} > \gamma) \tag{45}$$

All in all, the self-adaptive step size control method based on

corrector iterations according to [21] is used at the beginning of each predictor step. Because of more iterations in multi-area power flow calculations, a polynomial fitting method is used to accelerate the multi-area CPF when the deviation of the voltage or reactive power is large enough to generate effective nonlinear fitting parameters. It can also be seen that the main computational burden of the improved step size control method comes from the fitting parameters calculations in the three-dimensional matrix of each bus, which is much smaller than the power flow calculations in the CPF corrector.

3.5. Diagram of the computational procedure

The computational procedure is shown in Fig. 2, which illustrates the relationships among the predictor, corrector and step size control. First, the power flow and network data, such as tie line connections and AGC information, must be input. Then, the calculation parameters and variables including bus voltages, bus angles and area imbalance power are initialized. In the case of $\lambda = 0$, through the Newton iterative method-based calculations, the power flow solutions are obtained as the initialization for CPF. Since the CPF calculations alternate between the predictor and corrector, the predictor process begins with the identification of SNB ($d\lambda < 0$) or LIB ($\delta < 0$) based on the calculation results of the corrector. A sufficiently small step size is required for CPF to generate calculation results with high accuracy in the case of encountering SNB or LIB. Therefore, if the computation encounters bifurcation points in the case of a large step size, σ is set to σ_{min} and $\mathbf{X}$ is rolled back to $\mathbf{X}^{base}$, which has been recorded over time. Moreover, improved step size control is deployed at the end of the predictor step to obtain $\mathbf{X}^{pre}$ in accordance with the base point and step size. In addition, the variables are gradually corrected in the Newton iteration of the corrector to obtain the numerical power flow solution with respect to a certain value of λ. To address the numerical divergence of the Newton iteration process caused by an excessively large step size, a rollback scheme is deployed by the corrector. The improved step size control method displays high efficiency in generating suitable values of σ to adaptively accelerate and decelerate the predictor–corrector computational process.

4. Case studies

In this section, PTLs are analyzed in three case studies to verify the proposed methods. The conventional step size control method and the improved step size control method are adopted respectively in PTLs calculation. The conventional step size control method comes from [21].

Case 1 involves the IEEE 5-bus test system, in which the power grid is divided into two subareas. The simplicity of this case allows one to explore the detailed calculation process of the proposed CPF method.

The IEEE 118-bus test system deployed in Case 2 involves three subareas interconnected in series and is suitable for analyses of the critical limits for power transferred from the source area to the demand area through an intermediate area.

In case 3, the study of Polish 2383-bus test system which involves six subareas interconnected is introduced to verify the efficiency of proposed methods to analyze a real size power system.

Note that all the numerical values in the case studies are presented as per unit values with a base value of 100 MVA.

4.1. Case 1

As shown in Fig. 3, the IEEE 5-bus test system is divided into two control areas, and the measurement points for power exchange are labeled. Here, the PTLs from Bus 5 (area 2) to Bus 2 (area 1) and from Bus 4 (area 1) to Bus 1 (area 2) are separately analyzed using the proposed method.

Table 1 provides the information for the AGC system and the reactive power limitations of the two generation buses.

The calculation results for the PTL from Bus 4 to Bus 1 are shown in Fig. 4. With the synchronous growth in power generation at Bus 4 and the power demand at Bus 1, the reactive power outputs of the PV buses (Bus 4 and Bus 5) are increased, and the voltage of the PQ buses (Bus 1, Bus 2 and Bus 3) is decreased. When λ reaches approximately 0.558, the generator reactive power output at Bus 5 reaches the upper limit, which results in the bus mode conversion of Bus 5 and LIB, as confirmed by sensitivity analysis.

The sensitivity analysis results while the generator reactive power output at Bus 5 reaches the upper limit are shown in Table 2

In addition, the conventional and improved methods for step size control are separately employed and exhibit different performances, as shown in Fig. 5.

In this extreme and infrequent scenario, 20 predictor–corrector calculations are needed for CPF using the improved method, but 3564 calculations are required by the conventional method due to the continuous oscillation of step size at low levels. Then, CPF is used to analyse the PTL from Bus 5 to Bus 2, where λ reaches approximately 3.973 since LIB is encountered due to the reactive power constraint at Bus 5. Furthermore, the step sizes using the improved and conventional methods are compared in Fig. 6 in which the corresponding step counts are 465 and 545.

4.2. Case 2

In this case, an IEEE 118-bus test system, which consists of three

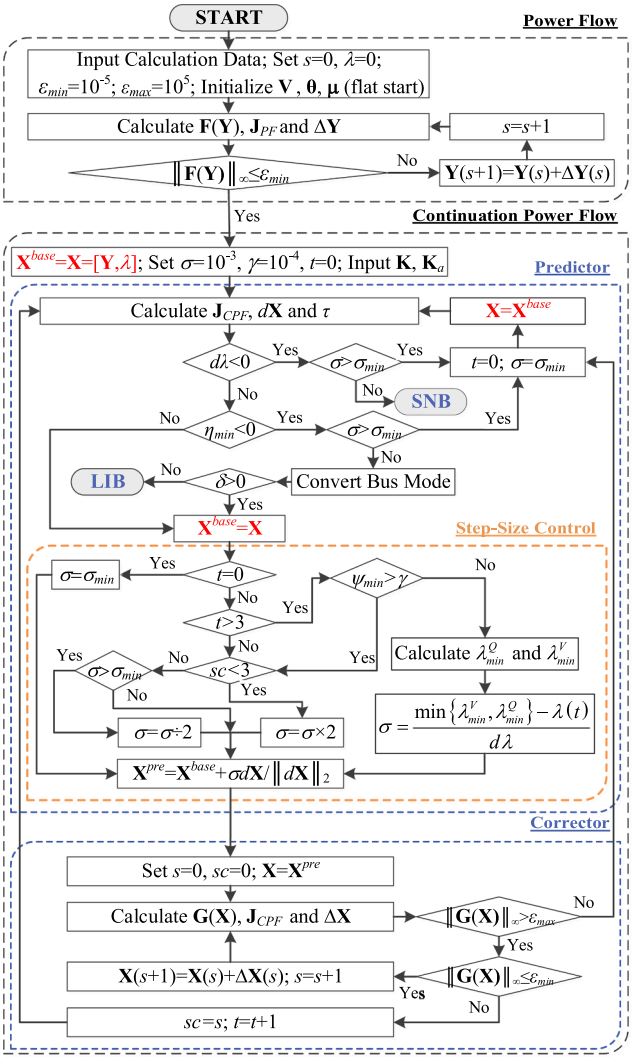


Fig. 2. Diagram of the CPF calculation process.

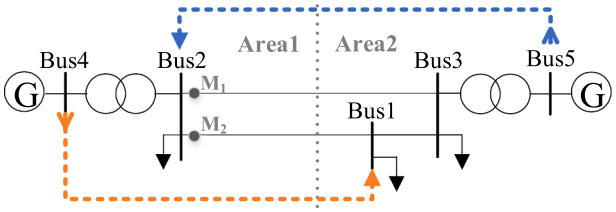


Fig. 3. Diagram of the IEEE 5-bus test system.

Table 1
Information for AGC.

Area	$\alpha_{i,a}$	Q^M	P_a^A
1	$\alpha_{4,1} = 1$	$Q_5^M = 3.0$	3.0005
2	$\alpha_{5,2} = 1$	$Q_4^M = 3.0$	−3.0005

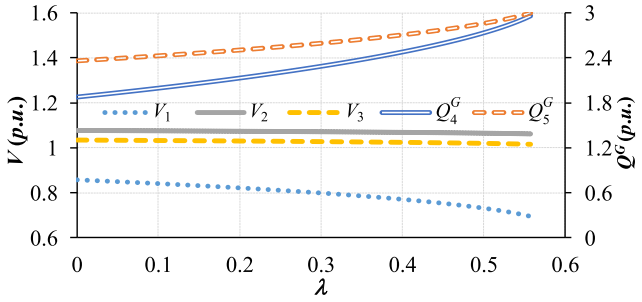


Fig. 4. PTL from Bus 4 to Bus 1 for the IEEE 5-bus system.

Table 2
Sensitivity analysis results of PTL from Bus 4 to Bus 1.

λ	Sensitivity analysis results				
	Bus-1	Bus-2	Bus-3	Bus-4	Bus-5
0.55688	−1.24528	−0.02406	−0.03935	PV	PV
0.55768	−1.28170	−0.02436	−0.03966	PV	PV
0.55775	−1.26422	−0.02438	−0.03968	PV	PV
0.5578	1.4889	0.005	0.2771	PV	PQ

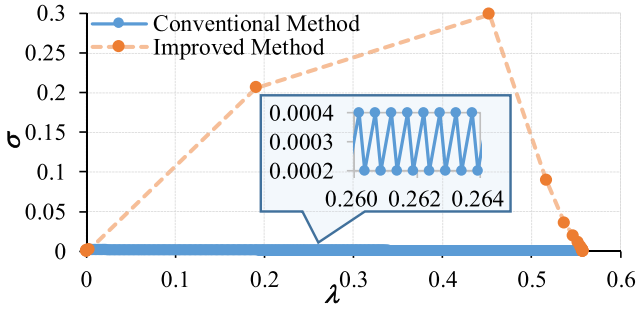


Fig. 5. Step size control method comparison for the PTL from Bus 4 to Bus 1.

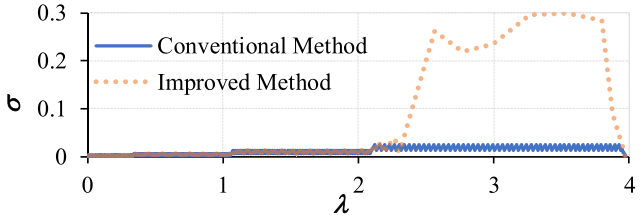


Fig. 6. Step size control method comparison for the PTL from Bus 5 to Bus 2.

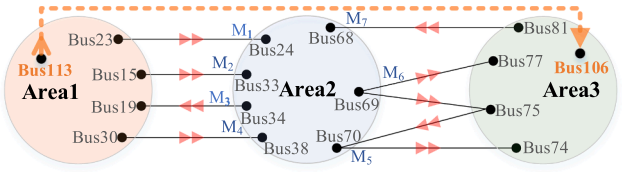


Fig. 7. Diagram of the IEEE 118-bus test system.

Table 3
AGC Parameters for the IEEE 118-Bus Test System.

Area	$\alpha_{i,a}$	P_a^A
1	$\alpha_{10,1} = 0.3$; $\alpha_{12,1} = 0.1$; $\alpha_{25,1} = 0.3$; $\alpha_{26,1} = 0.3$;	0.7481
2	$\alpha_{49,2} = 0.1$; $\alpha_{59,2} = 0.1$; $\alpha_{61,2} = 0.1$; $\alpha_{65,2} = 0.2$; $\alpha_{66,2} = 0.2$	0.6964
3	$\alpha_{80,3} = 0.3$; $\alpha_{89,3} = 0.5$; $\alpha_{100,3} = 0.2$	−1.4382

subareas, is analyzed using the presented CPF method. The detailed information for the tie lines, including the bus names on both sides of each line, the original power flow direction and the active power measurement positions, is given in Fig. 7, which also illustrates the point-to-point buses used for PTL testing.

The AGC-related participation coefficients and power exchange regulation information for different subareas can be found in Table 3.

Then, the proposed CPF is used to calculate the PTL from Bus 113 to Bus 106, for which the incremental source power in area 1 is transferred through area 2 to support the demand growth in area 3. The calculation results are shown in Fig. 8. As observed, as λ increases, five PV generation buses are converted in sequence to PQ mode due to the reactive power limitation, followed by decreases in the bus voltages. However, only the bus mode conversion for Bus 107 leads to LIB in the bus system state, and λ for the PTL reaches approximately 3.363. Moreover, the improved and conventional step size control methods are separately used for CPF, and 566 and 743 predictor–corrector steps are generated.

The variation in the step size for different step size methods is shown in Fig. 9, which illustrates the effectiveness of the improved method. Additionally, the improved method self-adaptively increases the step size and then automatically decreases the step size when the reactive power limit is approached at the corresponding generation buses.

4.3. Case 3

Fig. 10 shows the simplified representation of Polish 2383-bus test system, which consists of 6 subareas and 92 area tie-lines. Part of the information includes bus location, power exchange regulation and area lines about this system is shown in Table 4. Then the proposed CPF method is used to calculate the PTL from area 2 (Bus 654, Bus 712 and

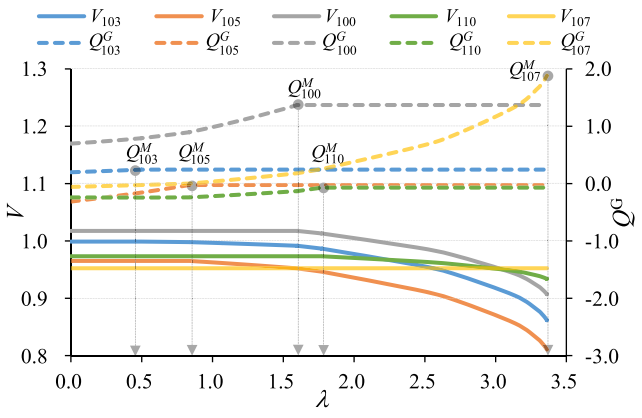


Fig. 8. PTL from Bus 113 to Bus 106 for the IEEE 118-bus system.

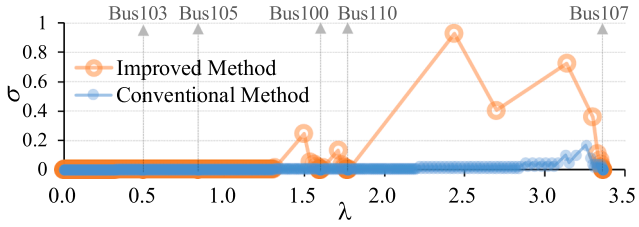


Fig. 9. Step size control method comparison for the 118-bus system.

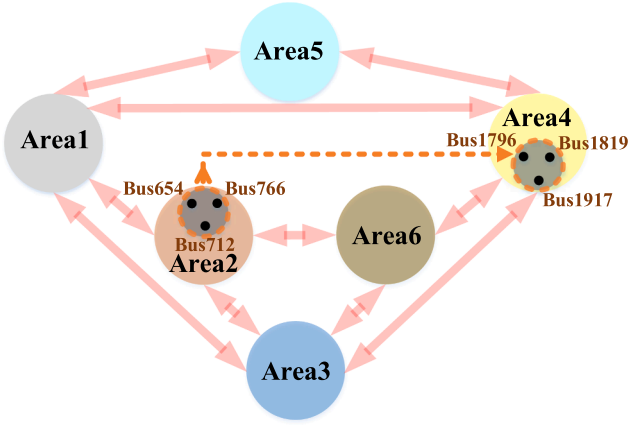


Fig. 10. Diagram of the Polish 2383-bus test system.

Table 4
Information for the Polish 2383-Bus Test System.

Area	Bus location	Area tie-lines number	P_a^A
1	1–22, 187–538	43	14.5606
2	23–55, 539–785	36	0.6385
3	56–110, 786–1601	38	17.667
4	111–154, 1602–2115	34	−9.7936
5	155–179, 2116–2381	20	−15.3263
6	180–186, 2382–2383	13	−7.7462

Table 5
Growth Coefficients of Polish 2383-Bus Test System.

Bus	Bus type	Growth coefficient			
		P_{gen}	Q_{gen}	P_{load}	Q_{load}
Bus 654	PV	1	*	0	0
Bus 712	PV	1	*	0	0
Bus 766	PV	1	*	0	0
Bus 1796	PQ	0	0	1	0
Bus 1819	PQ	0	0	1	0
Bus 1917	PQ	0	0	1	0

Bus 766) to area 4 (Bus 1796, Bus 1819 and Bus 1917). The growth coefficients of power generation and load demand are shown in Table 5. The calculation results are shown in Fig. 11. As observed in Fig. 11, the power output from area 2 and the power input to area 4 are increased as λ increases. Five PV generation buses are converted to PQ mode in sequence due to the reactive power limitation.

The voltage magnitudes at bus 1796, bus 1819 and bus 1917 decrease while the load demand growth. When λ reaches approximately 3.621, system state encounters SNB. The performances in case of conventional and improved step size control methods are compared and shown in Fig. 12.

The conventional and improved step-size control strategies are separately employed in the case studies of different systems, for which the convergence time, predictor–corrector step count, convergence

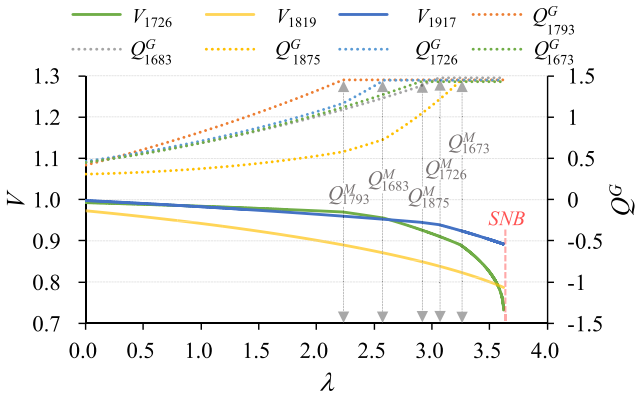


Fig. 11. PTL from Area 2 to Area 4 for the Polish 2383-bus system.

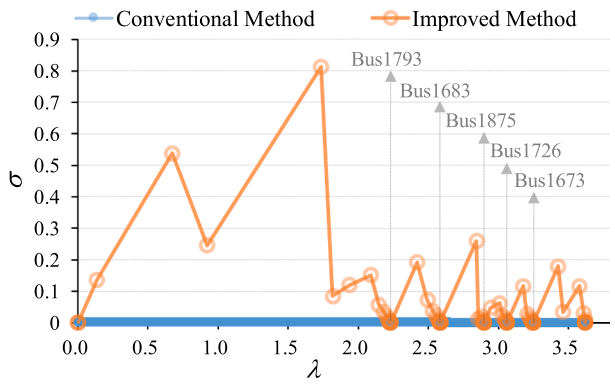


Fig. 12. Step size control method comparison for the 2383-bus system.

Table 6
Comparisons Between Different Method.

Test System		IEEE 5-Bus	IEEE 118-Bus	Polish 2383-Bus
Convergence time	Improved	0.11 s	5.34 s	3h46min
	Conventional	0.62 s	7.15 s	18h54min
Predictor-Corrector	Improved	20	566	143
	Conventional	3564	743	5278
Bifurcation point	Improved	0.5578	3.3626	3.6211
	Conventional	0.5578	3.3625	3.6211
Total step-size control time	Improved	7.2617 ms	6.6689 ms	103.7299 ms
	Conventional	1.1073 ms	0.8026 ms	5.8505 ms

accuracy and total step size control time are detailed in Table 6. Compared to convergence time, total step size control time is much smaller on improved step-size control strategies. The test computer is configured as follows. The CPU, memory and storage are Intel(R) Core (TM) i3-7100 @ 3.90 GHz Dual-Core Quad-threading, 12 GB 2400 MHz and Lenovo SSD ST610 120G. As shown in Table 6, the improved step size control method can significantly reduce computational burden while ensuring the convergence accuracy.

5. Conclusions

This study presents a new practical CPF model to calculate the PTL of power systems covering multi-area interconnected networks. Multi-area AGC operation is considered in the CPF model by including AGC participation coefficients, power balancing parameters and a formulation of the power exchange regulation among subareas. A polynomial fitting-based step size control method is proposed to enhance the efficiency and performance of predictor–corrector step size treatment in the CPF model. The proposed model and method use a detailed representation for the multi-area AGC to effectively generate the PTL

results. The case studies of IEEE 5-bus, IEEE118-bus and Polish 2383-bus test systems illustrate the convergence and effectiveness of the method. The comparisons of CPF calculations based on the proposed and conventional methods demonstrate the improvement in step size control obtained with the method proposed in this study.

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Conflict of interest statement

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Power Transfer Limit Calculation for Multi-area Interconnected Power Networks”.

Author Statement

Xiaoming Dong and Mengxia Wang discussed and conceived of the presented idea under the guidance of Peng Wang and Jinyu Wang. Xiaoming Dong and Chengfu Wang derived the formulations and performed the computations. Xiaoming Dong and Xupeng Hao design and written the programming code. Mengxia Wang, Chengfu Wang and Xiaoming Dong tested and verified the case. Xiaoming Dong, Xupeng Hao and Mengxia Wang summed up relevant data. Xiaoming Dong took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis, and manuscript. Suoying He and Xupeng Hao completed the response to the comments.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijepes.2020.105953.

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